

## BLAKELY, GA, USA

WMO: 720257

Lat: 31.397N

Lon: 84.895W

Elev: 65

StdP: 100.54

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 63835

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB  |             | Humidification DP/MCDB and HR |            |            |             |            |            | Coldest Month WS/MCDB |             |            |             | MCWS/PCWD to 99.6% DB |            | WSF          |
|---------------|-------------|-------------|-------------------------------|------------|------------|-------------|------------|------------|-----------------------|-------------|------------|-------------|-----------------------|------------|--------------|
|               |             |             | 99.6%                         |            |            | 99%         |            |            | 0.4%                  |             | 1%         |             |                       |            |              |
|               | 99.6%       | 99%         | DP                            | HR         | MCDB       | DP          | HR         | MCDB       | WS                    | MCDB        | WS         | MCDB        | MCWS                  | PCWD       |              |
| (a)<br>1      | (b)<br>-2.3 | (c)<br>-0.2 | (d)<br>-9.9                   | (e)<br>1.6 | (f)<br>3.5 | (g)<br>-7.3 | (h)<br>2.0 | (i)<br>4.0 | (j)<br>7.6            | (k)<br>13.7 | (l)<br>6.7 | (m)<br>12.4 | (n)<br>0.9            | (o)<br>340 | (p)<br>0.329 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 10.8                   | 36.2            | 24.7 | 35.0 | 24.6 | 33.8 | 24.4 | 27.5                | 31.1 | 27.0 | 31.0 | 26.3 | 30.8 | 2.1                  | 260  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 27.0                            | 22.9 | 29.0 | 26.1 | 21.6 | 28.8 | 25.1 | 20.4 | 28.4 | 87.5          | 31.4 | 85.1 | 31.2 | 82.2 | 30.7 | 30.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |         |         | Extreme Annual Temperature |          |                    |         | n-Year Return Period Values of Extreme Temperature |          |            |          |            |          |            |          |  |  |
|-------------------|---------|---------|----------------------------|----------|--------------------|---------|----------------------------------------------------|----------|------------|----------|------------|----------|------------|----------|--|--|
|                   |         |         | Mean                       |          | Standard Deviation |         | n=5 years                                          |          | n=10 years |          | n=20 years |          | n=50 years |          |  |  |
| 1%                | 2.5%    | 5%      | Min                        | Max      | Min                | Max     | Min                                                | Max      | Min        | Max      | Min        | Max      | Min        | Max      |  |  |
| (a) 6.0           | (b) 5.2 | (c) 4.6 | (e) -5.1                   | (f) 37.0 | (g) 2.5            | (h) 1.1 | (i) -6.9                                           | (j) 37.8 | (k) -8.4   | (l) 38.4 | (m) -9.7   | (n) 39.0 | (o) -11.5  | (p) 39.8 |  |  |
|                   |         |         | DB                         |          |                    |         |                                                    |          |            |          |            |          |            |          |  |  |
|                   |         |         | WB                         |          |                    |         |                                                    |          |            |          |            |          |            |          |  |  |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 19.7   | 10.3  | 12.3  | 15.3  | 19.2  | 23.3  | 27.1  | 27.9  | 27.7  | 25.3  | 20.5  | 14.2  | 12.4 |  |  |
|                                                                        | DBStd       | 7.35   | 5.50  | 4.84  | 4.57  | 3.30  | 3.07  | 1.74  | 1.75  | 1.95  | 2.47  | 4.10  | 4.43  | 4.99 |  |  |
|                                                                        | HDD10.0     | 143    | 63    | 29    | 9     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 11    | 31   |  |  |
|                                                                        | HDD18.3     | 936    | 254   | 177   | 115   | 28    | 4     | 0     | 0     | 0     | 1     | 26    | 137   | 194  |  |  |
|                                                                        | CDD10.0     | 3672   | 73    | 93    | 174   | 275   | 413   | 512   | 556   | 548   | 459   | 325   | 138   | 106  |  |  |
|                                                                        | CDD18.3     | 1422   | 6     | 7     | 22    | 53    | 158   | 262   | 297   | 290   | 210   | 93    | 14    | 10   |  |  |
|                                                                        | CDH23.3     | 14051  | 24    | 54    | 219   | 593   | 1595  | 2669  | 3139  | 2904  | 1849  | 828   | 124   | 50   |  |  |
| (14) Wind                                                              | WSAvg       | 1.3    | 1.6   | 1.6   | 1.7   | 1.5   | 1.3   | 1.1   | 1.0   | 0.9   | 1.2   | 1.2   | 1.2   | 1.4  |  |  |
|                                                                        | PrecAvg     | 1432   | 168   | 153   | 150   | 102   | 97    | 133   | 149   | 113   | 78    | 58    | 92    | 132  |  |  |
|                                                                        | PrecMax     | 2014   | 329   | 275   | 322   | 237   | 207   | 334   | 312   | 251   | 231   | 225   | 240   | 244  |  |  |
|                                                                        | PrecMin     | 943    | 25    | 23    | 2     | 11    | 8     | 21    | 30    | 43    | 6     | 0     | 22    | 27   |  |  |
|                                                                        | PrecStd     | 259    | 75    | 64    | 68    | 67    | 53    | 75    | 65    | 50    | 58    | 57    | 50    | 65   |  |  |
|                                                                        | DB          | 26.0   | 27.4  | 29.9  | 31.7  | 34.8  | 37.3  | 37.1  | 37.5  | 36.0  | 32.7  | 28.9  | 27.2  |      |  |  |
|                                                                        | MCWB        | 19.9   | 19.6  | 18.5  | 21.0  | 22.0  | 24.5  | 24.9  | 26.8  | 23.3  | 22.6  | 20.6  | 21.4  |      |  |  |
| (20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 2% DB       | 23.0   | 25.1  | 27.5  | 30.1  | 32.9  | 36.1  | 35.8  | 35.9  | 34.0  | 31.3  | 26.4  | 24.9  |      |  |  |
|                                                                        | 2% MCWB     | 18.6   | 17.8  | 18.8  | 20.5  | 22.3  | 24.2  | 25.2  | 25.5  | 23.3  | 22.0  | 19.2  | 20.1  |      |  |  |
|                                                                        | 5% DB       | 21.3   | 22.7  | 26.1  | 28.0  | 32.2  | 34.1  | 34.3  | 34.1  | 32.7  | 30.0  | 24.8  | 22.5  |      |  |  |
|                                                                        | 5% MCWB     | 18.0   | 17.0  | 17.9  | 19.7  | 22.0  | 24.2  | 25.2  | 25.4  | 23.1  | 21.5  | 18.7  | 19.7  |      |  |  |
|                                                                        | 10% DB      | 19.0   | 21.1  | 23.8  | 27.2  | 30.9  | 32.8  | 32.9  | 32.7  | 31.4  | 27.7  | 22.5  | 21.1  |      |  |  |
|                                                                        | 10% MCWB    | 16.4   | 16.4  | 16.9  | 18.9  | 21.9  | 23.9  | 25.1  | 25.2  | 23.2  | 20.8  | 17.7  | 18.6  |      |  |  |
|                                                                        | WB          | 21.7   | 21.3  | 22.0  | 23.9  | 25.9  | 27.5  | 28.1  | 29.0  | 27.1  | 26.1  | 23.9  | 23.0  |      |  |  |
| (28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4% MCDB   | 24.6   | 24.0  | 25.2  | 27.5  | 28.2  | 30.1  | 32.5  | 32.7  | 28.8  | 29.5  | 26.2  | 25.2  |      |  |  |
|                                                                        | 2% WB       | 20.4   | 19.9  | 20.7  | 22.9  | 24.7  | 26.5  | 27.3  | 27.9  | 25.9  | 24.9  | 22.0  | 21.7  |      |  |  |
|                                                                        | 2% MCDB     | 22.3   | 23.8  | 23.9  | 26.6  | 28.7  | 31.1  | 31.6  | 31.8  | 29.7  | 28.6  | 24.4  | 23.6  |      |  |  |
|                                                                        | 5% WB       | 18.8   | 18.8  | 19.8  | 22.0  | 23.9  | 25.8  | 26.7  | 27.2  | 25.3  | 23.6  | 20.6  | 20.5  |      |  |  |
|                                                                        | 5% MCDB     | 21.0   | 22.4  | 24.0  | 25.9  | 28.4  | 30.9  | 31.3  | 31.2  | 29.9  | 27.0  | 23.2  | 22.4  |      |  |  |
|                                                                        | 10% WB      | 17.0   | 17.3  | 18.7  | 20.9  | 23.2  | 25.3  | 26.1  | 26.5  | 24.6  | 22.4  | 18.9  | 18.6  |      |  |  |
|                                                                        | 10% MCDB    | 19.2   | 19.6  | 23.1  | 24.6  | 28.1  | 30.6  | 30.9  | 30.9  | 29.3  | 26.3  | 22.4  | 20.6  |      |  |  |
| (35) Mean Daily Temperature Range                                      | MDBR        | 11.4   | 12.6  | 13.4  | 13.7  | 12.9  | 11.5  | 10.8  | 10.6  | 11.3  | 12.9  | 13.4  | 11.5  |      |  |  |
|                                                                        | 5% DB MCDBR | 12.7   | 14.6  | 14.7  | 15.2  | 14.5  | 13.7  | 11.8  | 12.0  | 13.2  | 13.7  | 15.0  | 12.3  |      |  |  |
|                                                                        | 5% DB MCWBR | 8.0    | 8.4   | 7.2   | 6.6   | 5.4   | 4.2   | 3.8   | 4.0   | 4.5   | 5.6   | 8.1   | 8.5   |      |  |  |
|                                                                        | 5% WB MCDBR | 10.4   | 11.8  | 11.5  | 11.5  | 11.4  | 11.2  | 10.4  | 10.6  | 10.5  | 10.4  | 10.7  | 10.2  |      |  |  |
|                                                                        | 5% WB MCWBR | 7.8    | 8.4   | 6.7   | 6.6   | 5.1   | 4.3   | 4.2   | 4.2   | 4.4   | 4.9   | 7.2   | 7.9   |      |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        | 0.326  | 0.333 | 0.355 | 0.384 | 0.434 | 0.477 | 0.509 | 0.500 | 0.426 | 0.373 | 0.351 | 0.331 |      |  |  |
|                                                                        | taud        | 2.515  | 2.489 | 2.440 | 2.362 | 2.252 | 2.196 | 2.116 | 2.139 | 2.328 | 2.450 | 2.455 | 2.516 |      |  |  |
|                                                                        | Ebn at Noon | 903    | 928   | 928   | 908   | 860   | 819   | 790   | 794   | 844   | 871   | 863   | 876   |      |  |  |
|                                                                        | Edh at Noon | 87     | 98    | 110   | 124   | 139   | 147   | 158   | 153   | 122   | 100   | 92    | 83    |      |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      | 3.00   | 3.57  | 4.77  | 5.86  | 6.43  | 6.10  | 6.02  | 5.55  | 4.98  | 4.30  | 3.37  | 2.61  |      |  |  |
|                                                                        | RadStd      | 0.18   | 0.43  | 0.39  | 0.47  | 0.45  | 0.45  | 0.36  | 0.37  | 0.40  | 0.45  | 0.28  | 0.26  |      |  |  |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (42 neighbors) |  | +0.46 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | +160    | +104    |

Nomenclature: See separate page